

# "These are hard questions:" An Investigation of the Developmental Abilities of 6th Graders to Answer Survey Items

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## Background

- Data collected about health behaviors, attitudes about school, and teacher opinions often come from adolescents
- Many resources are invested in education and health research focused on youth
- It is important to test procedures and use different methods to improve the quality of responses to survey questions

## Research Questions

1) Do survey procedures work as planned in the time allotted during an actual school setting?
2) Can middle school students understand and accurately report on questionnaires about behaviors, attitudes and opinions?

## Methods

### Sample

9 middle school students from Sojourner Truth Academy (5 males, 4 females)

### Procedures

#### *Pilot Survey*

105 questions on 12 total pages, separated into different sections including:

- ❖ School experiences
- ❖ Beliefs and attitudes
- ❖ Family and friends
- ❖ Alcohol and drugs

#### *Cognitive Interview*

41 questions with prompts and probes to measure student comprehension, retrieval, and response abilities

### Analysis

Qualitative analysis of observations & staff notes

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## Results

**Figure 1: Sample Characteristics**

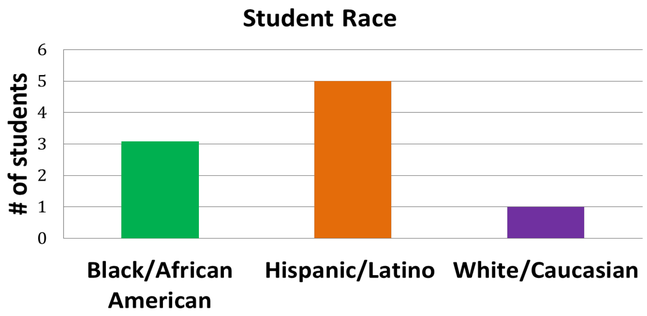



**Figure 2: Example of Response Assessment**

14. Please rate your agreement with each statement.

| | Strongly Agree | Agree | Disagree | Strongly Disagree |
|---|---|---|---|---|
| Option 1. | | | | |
| A. Overall, adults at my school treat students fairly. | ☐ | ☐ | ☐ | ☐ |
| B. The tests in my classes do a good job of measuring what I'm able to do. | ☐ | ☐ | ☐ | ☐ |
| C. In my school, students are given a chance to help make decisions | ☐ | ☐ | ☐ | ☐ |
| Option 2. | YES! | yes | no | NO! |
| A. Overall, adults at my school treat students fairly. | ☐ | ☐ | ☐ | ☐ |
| B. The tests in my classes do a good job of measuring what I'm able to do. | ☐ | ☐ | ☐ | ☐ |
| C. In my school, students are given a chance to help make decisions | ☐ | ☐ | ☐ | ☐ |
| Option 3. | Definitely Yes | Yes | No | Definitely No |
| A. Overall, adults at my school treat students fairly. | ☐ | ☐ | ☐ | ☐ |
| B. The tests in my classes do a good job of measuring what I'm able to do. | ☐ | ☐ | ☐ | ☐ |
| C. In my school, students are given a chance to help make decisions | ☐ | ☐ | ☐ | ☐ |

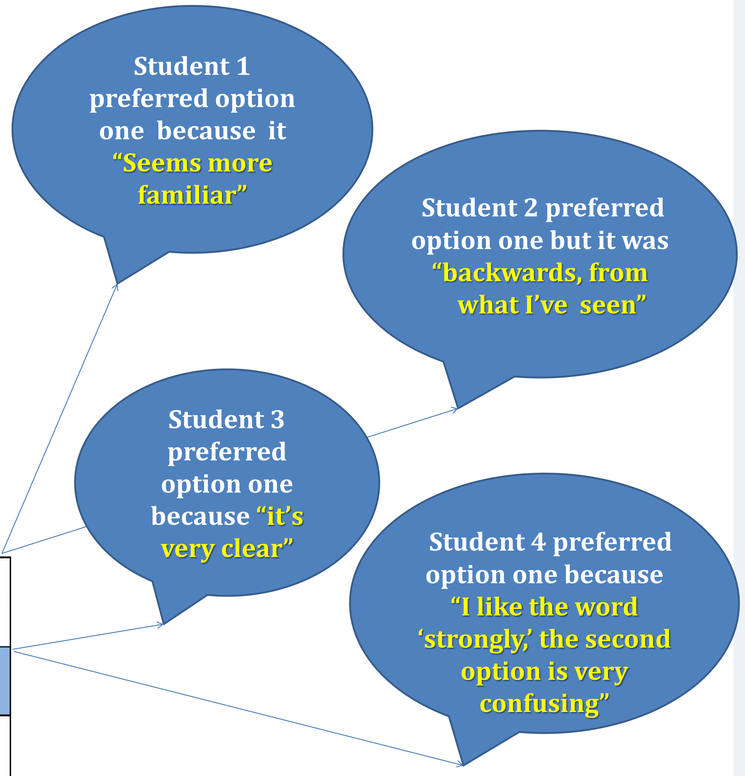



**Figure 3: Example of Comprehension Assessment**

**9. How many adults do you live with who provide care for you (like a parent)?**

☐ 0 ☐ 3
☐ 1 ☐ 4
☐ 2 ☐ 5 or more

**Follow up question:**
**'What does this question mean to you?"**

**Student 1 response**
*"it might be hard to answer if you're in foster care" adding… "if mom and 18 year old sister= 2, but if mom and 15 year old sister = 1"*

**Student 2 response**
*"mom, cousin, and dad provide care for me"*

**Student 3 response**
*"mom, dad, mom's friend, and uncle provide care for me at home"*

**Student 4 response**
*"Means people living with you who do stuff for you. I'd check 5 (parents and my older brothers). Brothers cook, protect me, be sure I don't get in trouble."*

## Discussion

**Pilot Survey Findings:**

- 28.5 minutes to complete
- Took longer than a typical advisory period (20 minutes)
- Regular class period is 40-50 minutes
- Length of survey creates added burden for study participants & classroom teachers who give up class time

**Cognitive Interview Findings:**

- Students comprehended questions
- Retrieved certain memories to answer items
- Picked responses that demonstrated their understanding of certain questions

**Relation to Other Studies**

- These findings are similar to those found in the Elementary School Success Profile study (Woolley & Bowen, 2006):
  - ❖ Most students could read questions on the cognitive interview guide without any help
  - ❖ Some rewording is needed to improve age appropriateness of questions

**Limitations**

- Sample size

**Importance/Next Steps**

- Results will guide revisions for carrying out a longitudinal study, Partnering for Healthy Student Outcomes, to improve academic and health outcomes for students in diverse, urban middle schools
  - ❖ Revise plan for scheduling surveys in regular classrooms
  - ❖ Reword some questions to accommodate for middle school cognitive abilities

## Implications

- The research team was able to refine study procedures to increase the quality of survey data used to measure academic and health outcomes for middle school students

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